



SustainAI 2026

Intelligence for a Sustainable Tomorrow



Innovate



Collaborate



Impact

INDUSTRY ACADEMIA CONCLAVE 2.0



IDEATHON

— AI FOR A SUSTAINABLE TOMORROW —



REUSEGRID AI

Team PUI
(PowerUp India)

TEAM MEMBERS

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Track: AI for SDGs (SDG 7 & 9)

INSTITUTION

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The Problem Statement



Current Landscape

- Used EV and electronic batteries are increasing, while many still contain usable cells and components.
- Discarding whole batteries creates e-waste and wastes valuable materials that could be reused.
- Clinics, schools, communities and microgrids need affordable and reliable clean-energy storage.



The Core Challenge

- We need a safe way to identify which used batteries and components can still serve a useful purpose.
- Repairable and reusable parts should be recovered instead of sending the whole battery to waste.
- Recovered storage should be matched to places where reliable, affordable and clean energy is most needed.



Our Solution

A dual-pipeline circular energy platform combining edge-based multi-battery diagnostics with component recovery, repair, and AI-optimized reconstruction for underserved grids.



Multi-Battery & Safety Gate

Collects data from multiple used batteries and checks their voltage, current, temperature and condition.
Unsafe batteries are rejected before reuse.



Component Pool & Repair

Identifies repairable and reusable parts, groups safe components from different batteries, and checks whether they can work together safely.



AI Optimization & Matching

AI selects the best compatible components to create a second-life battery, then matches it with clinics, schools, homes or microgrids that need reliable clean energy.



SDG Alignment & Impact



Environmental

Life Extension: Adds 4–6 yrs to packs;

- recovers 85% viable modules.

Carbon Cut: Saves ~65 kg CO₂e/kWh vs

- new LFP manufacturing.

Ensures universal access to affordable, reliable, sustainable, and modern clean energy for all.



Social

Clinic Resiliency: 24/7 backup for

- vaccine storage & primary health labs.

Energy Equity: Low-cost clean storage

- for off-grid schools & microgrids.

Ensures sustainable consumption and production patterns to reduce waste



Economic

- **50% CapEx Savings:** Deployed at \$28–\$38/kWh vs \$70–\$90/kWh new storage.

- **EPR Revenue:** Tradable circular compliance credits under BWMR 2022.
Builds resilient infrastructure, sustainable industrialization, and fosters innovation.



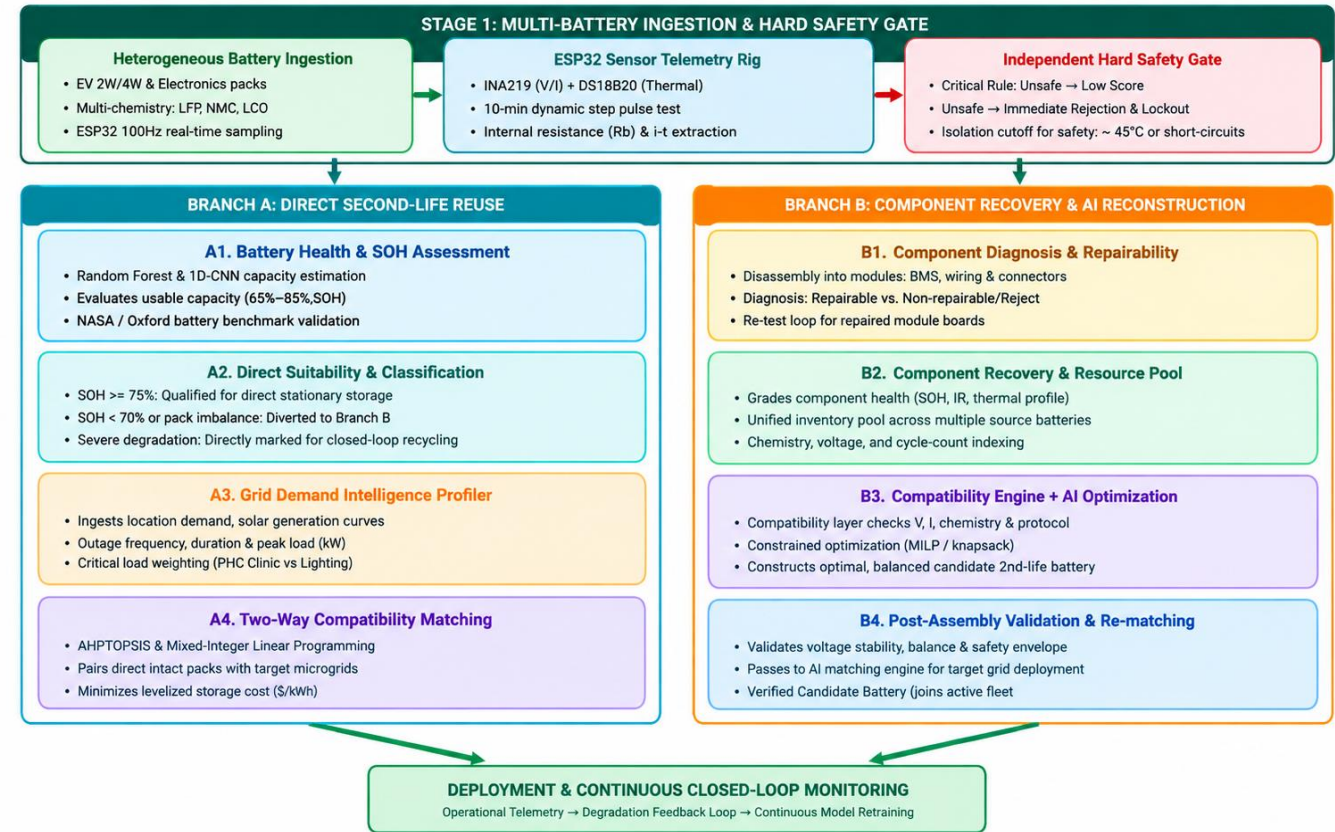
Architecture & Tech Stack

Core Technologies

- **Frontend:** Streamlit & Altair for the interactive dashboard, coupled with Cloudflare Tunnel for live public HTTPS access
- **Backend:** Python using Scikit-Learn (Random Forest) for health prediction and Google OR-Tools for allocation optimization.
- **Dataset:** Pandas & OpenPyXL managing the 500-battery benchmark and 800-component CSV/XLSX databases.
- **Hardware:** ESP32 microcontroller (Arduino C++) with INA219 (V/I) and DS18B20 (Temp) sensors connected via PySerial.

REUSEGRID AI — DUAL-PIPELINE CIRCULAR ARCHITECTURE

Integrated Direct Battery Allocation & Component-Level Recovery, Repair & AI Reconstruction





Implementation Roadmap

1

1. Prototype

- ESP32 hardware test rig with INA219 & DS18B20.
- 10-min pulse test & rapid SOH regression model.
- Streamlit dual-pipeline MVP with 500-battery dataset.

2

2. Validation

- Benchmark on NASA & Oxford battery datasets.
- Component recovery & repair testing loop.
- Hard safety cutoff ($<45^{\circ}\text{C}$) & active BMS balancing.

3

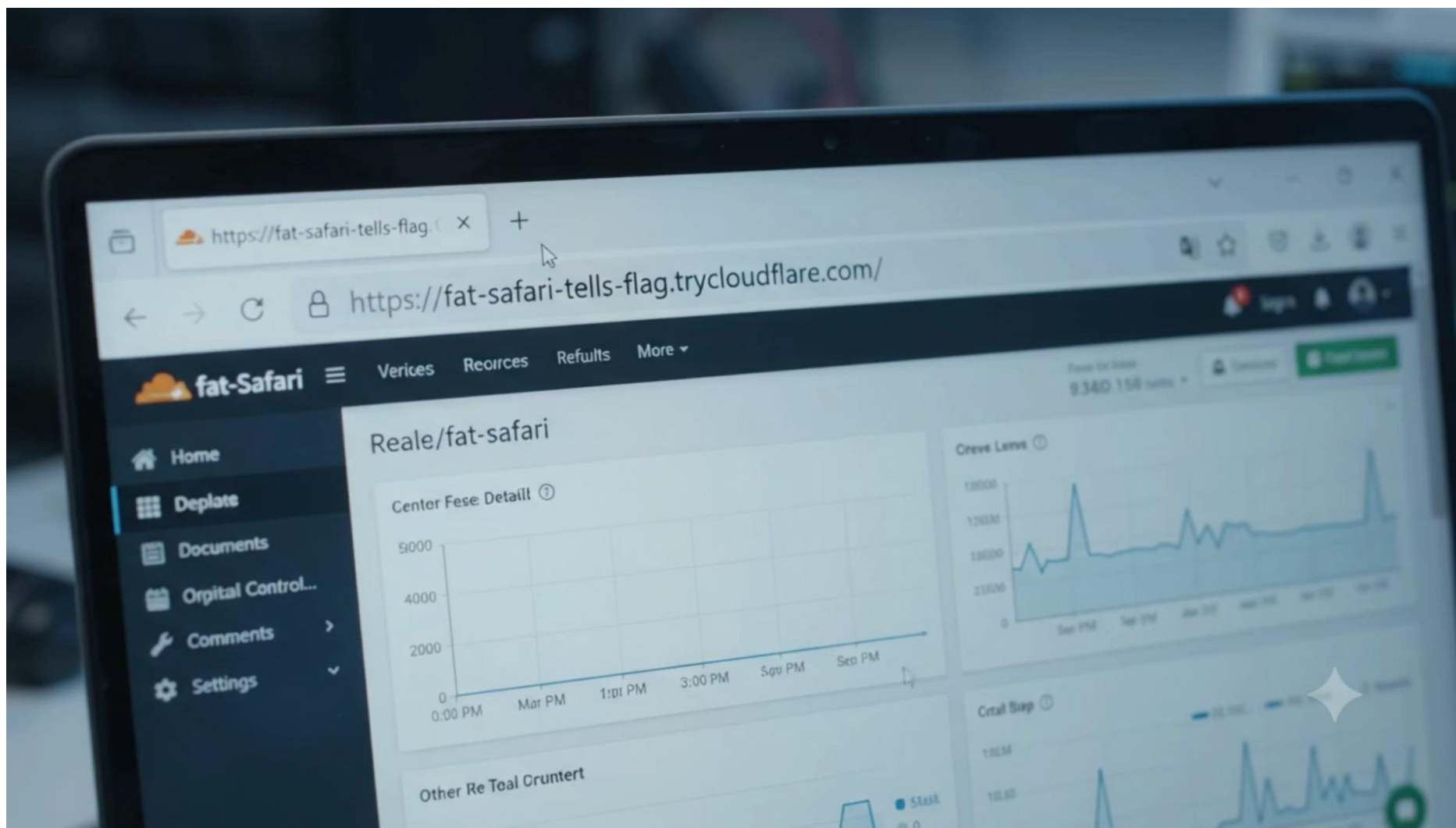
3. Pilot Launch

- 2–5 kWh reconstructed candidate pack at a rural PHC.
- Fleet sourcing partnership with 2W EV aggregators.
- Closed-loop degradation feedback into dataset.

4

4. Scale & EPR

- Cloud SaaS component inventory & matching.
- QR Digital Battery Passports for full traceability.
- Direct integration with India's EPR regulatory portal.





Thank You!

"Empowering sustainable tomorrows by transforming yesterday's electric mobility into tomorrow's resilient community grids."

CONTACT & LINKS



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<https://github.com/shreyasahu1234/Power-Up-India-PUI->